PRACTICAL COURSE WORKBOOK

Applied Machine Learning

Turn untidy data into a defensible finding

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a cleaned dataset, analysis and decision-ready chart
SKILLS	Scikit-learn, Regression, Classification, Clustering, Critical evaluation, Documentation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use Scikit-learn appropriately in a realistic, bounded task.
- Use Regression appropriately in a realistic, bounded task.
- Use Classification appropriately in a realistic, bounded task.
- Use Clustering appropriately in a realistic, bounded task.

WHO THIS IS FOR

Learners using data and code to answer practical questions.

BEFORE YOU BEGIN

- Basic numeracy and file-management skills

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Applied Machine Learning is a structured, practical course covering Scikit-learn, Regression, Classification, Clustering. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Scikit-learn: Data foundations	1. Problem framing with Scikit-learn 2. Data types with Regression 3. Tool setup with Classification
02 Regression: Analysis	1. Cleaning with Clustering 2. Exploration with Scikit-learn 3. Modelling with Regression
03 Classification: Evaluation	1. Validation with Classification 2. Uncertainty with Clustering 3. Communication with Scikit-learn
04 Clustering: Applied project	1. Workflow with Regression 2. Review with Classification 3. Reproducibility with Clustering

HOW TO USE EACH LESSON PAGE

1. Read the objective and identify the decision it supports.
2. Attempt the retrieval prompt before checking the online explanation.
3. Reproduce the worked example with the stated input.
4. Test one normal case and one boundary or failure case.
5. Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

Scikit-learn: Data foundations

Apply Scikit-learn through problem framing, data types, tool setup.

LESSON 01 / 18 MIN

Problem framing with Scikit-learn

Learning objectives

- Explain problem framing in the context of Applied Machine Learning.
- Apply Scikit-learn to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

problem framing, Scikit-learn, data

Retrieval prompt

In Applied Machine Learning, which evidence best supports a problem framing result produced with Scikit-learn?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 02 / 22 MIN

Data types with Regression

Learning objectives

- Explain data types in the context of Applied Machine Learning.
- Apply Regression to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

data types, Regression, data

Retrieval prompt

In Applied Machine Learning, which evidence best supports a data types result produced with Regression?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 03 / 26 MIN

Tool setup with Classification

Learning objectives

- Explain tool setup in the context of Applied Machine Learning.
- Apply Classification to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

tool setup, Classification, data

Retrieval prompt

In Applied Machine Learning, which evidence best supports a tool setup result produced with Classification?

Evidence to keep [ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

Regression: Analysis

Apply Regression through cleaning, exploration, modelling.

LESSON 04 / 30 MIN

Cleaning with Clustering

Learning objectives

- Explain cleaning in the context of Applied Machine Learning.
- Apply Clustering to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

cleaning, Clustering, data

Retrieval prompt

In Applied Machine Learning, which evidence best supports a cleaning result produced with Clustering?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 05 / 18 MIN

Exploration with Scikit-learn

Learning objectives

- Explain exploration in the context of Applied Machine Learning.
- Apply Scikit-learn to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

exploration, Scikit-learn, data

Retrieval prompt

In Applied Machine Learning, which evidence best supports a exploration result produced with Scikit-learn?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 06 / 22 MIN

Modelling with Regression

Learning objectives

- Explain modelling in the context of Applied Machine Learning.
- Apply Regression to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

modelling, Regression, data

Retrieval prompt

In Applied Machine Learning, which evidence best supports a modelling result produced with Regression?

Evidence to keep [ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

Classification: Evaluation

Apply Classification through validation, uncertainty, communication.

LESSON 07 / 26 MIN

Validation with Classification

Learning objectives

- Explain validation in the context of Applied Machine Learning.
- Apply Classification to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

validation, Classification, data

Retrieval prompt

In Applied Machine Learning, which evidence best supports a validation result produced with Classification?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 08 / 30 MIN

Uncertainty with Clustering

Learning objectives

- Explain uncertainty in the context of Applied Machine Learning.
- Apply Clustering to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

uncertainty, Clustering, data

Retrieval prompt

In Applied Machine Learning, which evidence best supports a uncertainty result produced with Clustering?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 09 / 18 MIN

Communication with Scikit-learn

Learning objectives

- Explain communication in the context of Applied Machine Learning.
- Apply Scikit-learn to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

communication, Scikit-learn, data

Retrieval prompt

In Applied Machine Learning, which evidence best supports a communication result produced with Scikit-learn?

Evidence to keep [ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

Clustering: Applied project

Apply Clustering through workflow, review, reproducibility.

LESSON 10 / 22 MIN

Workflow with Regression

Learning objectives

- Explain workflow in the context of Applied Machine Learning.
- Apply Regression to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

workflow, Regression, data

Retrieval prompt

In Applied Machine Learning, which evidence best supports a workflow result produced with Regression?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 11 / 26 MIN

Review with Classification

Learning objectives

- Explain review in the context of Applied Machine Learning.
- Apply Classification to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

review, Classification, data

Retrieval prompt

In Applied Machine Learning, which evidence best supports a review result produced with Classification?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 12 / 30 MIN

Reproducibility with Clustering

Learning objectives

- Explain reproducibility in the context of Applied Machine Learning.
- Apply Clustering to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

reproducibility, Clustering, data

Retrieval prompt

In Applied Machine Learning, which evidence best supports a reproducibility result produced with Clustering?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable Applied Machine Learning project using Scikit-learn, Regression, Classification.

DELIVERABLE	A working Applied Machine Learning artefact plus an evidence-based self-review.
TOOLS	A suitable Scikit-learn environment, A plain-text decision log, Test data or realistic sample material

Production plan

1. Define the intended user, outcome and constraints.
2. Create the smallest complete result using Scikit-learn.
3. Test one normal case, one boundary case and one failure response.
4. Revise the work from the evidence and preserve before-and-after results.
5. Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER
State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS
List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE
Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE
What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION
What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION
Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

The Python Tutorial

Python Software Foundation - reviewed 2026-08-21

https://docs.python.org/3/tutorial/

User Guide

scikit-learn - reviewed 2026-08-21

https://scikit-learn.org/stable/user_guide.html

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Scikit-learn scope.